

Module 4: Vector Spaces

4.1

Def. of a Group: Let G be a non-empty

set together with one binary operation ' $\circ$ '. Then $(G, \circ)$ is called a group if it satisfies the following:

(i) Associativity: $a \circ (b \circ c) = (a \circ b) \circ c$
for $a, b, c \in G$,

(ii) Existence of identity: For all $a \in G$,

there exists an element $e \in G$ such that $a \circ e = e \circ a = a$.

Here e is called an identity element of G with respect to ' $\circ$ '.

(iii) Existence of inverse: For each $a \in G$,

there exists an element $b \in G$ such that $a \circ b = b \circ a = e$.

Here b is called an inverse of a in G with respect to ' $\circ$ '.

Note: A mapping $\circ : G \times G \rightarrow G$ is called a binary operation on G .

E.g. (i) $(\mathbb{N}, +)$ is not a group.

for, $0 \notin \mathbb{N}$ (0 is an additive identity.)

(ii) $(\mathbb{Z}, +)$ is a group.

(iii) $(\mathbb{Z}, \cdot)$ is not a group.

for, $2 \in \mathbb{Z}$ but $\frac{1}{2} \notin \mathbb{Z}$

i.e., multiplicative inverse of 2 is not in $\mathbb{Z}$.

(iv) $(\mathbb{R}, +)$ is a group.

(v) $(\mathbb{R}, \cdot)$ is not a group.

for, $0 \in \mathbb{R}$ has no multiplicative inverse.

(vi) $(\mathbb{R}^*, \cdot)$ is a group, where $\mathbb{R}^* = \mathbb{R} - \{0\}$.

Abelian Group: A group $(G, \cdot)$ is said to

be abelian if $a \cdot b = b \cdot a$ for all $a, b \in G$.

e.g. (i) $(\mathbb{Z}, +)$, $(\mathbb{R}, +)$, $(\mathbb{Q}, +)$ are all abelian groups.

(ii) $(\mathbb{R}^*, \cdot)$ is an abelian group.

Def. of a Ring: Let R be a non-empty set together with two binary operations '+' and '•'. Then $(R, +, \cdot)$ is called a ring if

- (i) $(R, +)$ is an abelian group
- (ii) $(R, \cdot)$ is a semi group
(i.e., $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all $a, b, c \in R$)
- (iii) Distributive laws hold good in R .
i.e., $a \cdot (b + c) = a \cdot b + a \cdot c$
and $(a + b) \cdot c = a \cdot c + b \cdot c$
for all $a, b, c \in R$.

e.g.:

- (i) $(\mathbb{Z}, +, \cdot)$ is a ring
- (ii) $(\mathbb{R}, +, \cdot)$ is a ring
- (iii) $(\mathbb{Q}, +, \cdot)$, $(\mathbb{C}, +, \cdot)$ are all rings.

Definition of a Field: Let F be a non-empty set with two binary operations '+' and '•'. Then $(F, +, \cdot)$ is called a field if

- (i) $(F, +)$ is an abelian group
- (ii) $(F^*, \cdot)$ is an abelian group ($F^* = F - \{0\}$)
- (iii) Distributive laws are hold good in F

i.e., $a \cdot (b+c) = a \cdot b + a \cdot c$
and $(a+b) \cdot c = a \cdot c + b \cdot c$ for all $a, b, c \in F$.

e.g. (i) $(\mathbb{Z}, +, \cdot)$ is not a field.

(ii) $(\mathbb{Q}, +, \cdot)$ is a field

(iii) $(\mathbb{R}, +, \cdot)$ and $(\mathbb{C}, +, \cdot)$ are fields.

Vector Space: Let V be a non-empty set whose elements are called vectors and F be a field whose elements are called scalars. Then $(V, +, \cdot)$ is said to be a vector space over F if

(i) $(V, +)$ is an abelian group.

(ii) $a \in F, \alpha \in V \Rightarrow a \cdot \alpha \in V$ ($\forall a \in F, \alpha \in V$)

(iii) $a \cdot (\alpha + \beta) = a \cdot \alpha + a \cdot \beta$ for $a \in F$ and $\alpha, \beta \in V$

(iv) $(a+b) \cdot \alpha = a \cdot \alpha + b \cdot \alpha$ for $a, b \in F$ and $\alpha \in V$

(v) $a \cdot (b \cdot \alpha) = (a \cdot b) \cdot \alpha$ for $a, b \in F$ and $\alpha \in V$

and (vi) $1 \cdot \alpha = \alpha$ for $1 \in F$ and $\alpha \in V$.

Note:

If $(V, +, \cdot)$ is a vector space over a field F , then we say $V(F)$ is a vector space (or, a linear space)

eg. (i) $\mathbb{R}$ is a vector space over a field $(\mathbb{Q}, +, \cdot)$
i.e., $\mathbb{R}(\mathbb{Q})$ is a vector space

(ii) $\mathbb{C}(\mathbb{R})$ is a vector space

(iii) $\mathbb{R}(\mathbb{C})$ is not a vector space.

for, $1 \in \mathbb{R}$ and $i \in \mathbb{C} \Rightarrow 1 \cdot i = i \notin \mathbb{R}$

(iv) $\mathbb{R}^n(\mathbb{R})$ is a vector space ($n=1, 2, 3, \dots$)

(v) $(M_{2 \times 2}, +, \cdot)$ is a vector space

over a field $(\mathbb{R}, +, \cdot)$ (here
 $M_{2 \times 2} = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid a, b, c, d \in \mathbb{R} \right\}$)

Subspace: Let $V(F)$ be a vector space.

A non-empty subset W of V is ~~called~~
said to be a subspace of V over a field F if

W is itself a vector space over F under
the same operations as defined on V .

Example ① Clearly

$W = \{ (a, b, c) \mid a, b, c \in \mathbb{R} \}$ is a subspace
of a vector space $\mathbb{R}^3(\mathbb{R})$.

② $W = \left\{ \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \mid a, b \in \mathbb{R} \right\}$ is a
a vector space

Subspace of $M_{2 \times 2}(\mathbb{R})$.

* The necessary and sufficient condition for a non-empty subset W of a vector space $V(F)$ to be a subspace of V is

$$a, b \in F \text{ and } \alpha, \beta \in W \Rightarrow a\alpha + b\beta \in W.$$

Note: ① The intersection of any two subspaces of a vector space is a subspace.

② The intersection of an arbitrary collection of subspaces of a vector space is also a subspace.

③ The union of two subspaces of a vector space is not necessarily a subspace.

④ The union of two subspaces W_1 and W_2 of a vector space $V(F)$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Def.: Let W_1 and W_2 be two subspaces of a vector space $V(F)$, then $W_1 + W_2 = \{ \alpha + \beta \mid \alpha \in W_1, \beta \in W_2 \}$ is called the linear sum of W_1 and W_2 .

Note: $W_1 + W_2$ is a subspace of $V(F)$.

Def.: Let W_1 and W_2 be two subspaces of a vector space $V(F)$. Then V is said to be a

direct sum of w_1 and w_2 if each element of V can be uniquely expressed as the sum of an element of w_1 and an element of w_2 .

If V is a direct sum of w_1 and w_2 , then we write $V = w_1 \oplus w_2$

* The necessary and sufficient condition for a vector space V to be the direct sum of its subspaces w_1 and w_2 are

$$(i) V = w_1 + w_2 \quad (ii) w_1 \cap w_2 = \{0\}.$$

Linear combination of vectors

Def.: Let $V(F)$ be a vector space. An element $\alpha \in V$ is said to be a linear combination of $\alpha_1, \alpha_2, \dots, \alpha_n \in V$ if there exists $a_1, a_2, \dots, a_n \in F$ such that $\alpha = a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_n \alpha_n$.

Example: ① The vector $\alpha = (-7, -6)$ is a linear combination of the vectors $\alpha_1 = (-2, 3)$

and $\alpha_2 = (1, 4)$ $\left(\begin{array}{l} \text{Since } \alpha \in \text{span}\{\alpha_1, \alpha_2\} \\ \text{then} \end{array} \right)$

$$\text{since } (-7, -6) = 2(-2, 3) - 3(1, 4)$$

$$\text{i.e., } \alpha = 2\alpha_1 + (-3)\alpha_2$$

Linear Span: Let $V(F)$ be a vector space and S be any non-empty subset of V . Then the linear span of S (or, simply span of S) is denoted by $L(S)$ (or, $\text{span}(S)$) and defined as the set of all finite linear combinations of elements of S .

$$\text{i.e., } L(S) = \left\{ \sum_{i=1}^n a_i \alpha_i \mid a_i \in F, \alpha_i \in S \text{ for } i=1,2,\dots,n \right\}.$$

Clearly $S \subseteq L(S)$.

Note: ① $\text{span}(S)$ is the smallest subspace of V that contains S .

Examples: ① $L(\{(1,0,0), (0,1,0), (0,0,1)\})$
 $= \mathbb{R}^3(\mathbb{R}).$

for, for $(x,y,z) \in \mathbb{R}^3$, $(x,y,z) = x(1,0,0) + y(0,1,0) + z(0,0,1).$

② $L(\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\})$
 $= M_{2 \times 2}(\mathbb{R}).$

for, for $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}$, $\begin{bmatrix} a & b \\ c & d \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$

③ Show that the set $S = \{(2, 5), (1, 3)\}$ is a spanning set of $\mathbb{R}^2$ ($\mathbb{R}$)

sol: let $(a, b) \in \mathbb{R}^2$. Suppose $(a, b) = r(2, 5) + s(1, 3)$

$$\text{Then } 2r + s = a$$

$$5r + 3s = b$$

$$\text{i.e., } \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} r \\ s \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\xrightarrow{2R_2 - 5R_1} \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r \\ s \end{bmatrix} = \begin{bmatrix} a \\ 2b - 5a \end{bmatrix}$$

$$\text{which gives } 2r + s = a$$

$$\text{and } s = 2b - 5a$$

$$\Rightarrow r = \frac{6a - 2b}{2} \Rightarrow r = 3a - b$$

$$\therefore (a, b) = (3a - b)(2, 5) + (2b - 5a)(1, 3).$$

and hence every element of $\mathbb{R}^2$ can be

expressed ~~as a~~ linear combination of

~~elements~~ of S . Thus $L(S) = \mathbb{R}^2$

Linearly dependent and independent set of vectors.

Def: Let $V(F)$ be a vector space over a field F .

Then a finite set $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ of vectors in V is said to be (i) linearly dependent if there exists scalars $a_1, a_2, \dots, a_n \in F$

not all are zero such that

$$a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n = 0$$

(ii) linearly independent if $a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n = 0$
where $a_1, a_2, \dots, a_n \in F$
implies $a_1 = 0, a_2 = 0, \dots, a_n = 0$

Example: ① The set of vectors $(2, 1, 4), (1, -1, 2), (3, 1, -2)$ i.e., $\{(2, 1, 4), (1, -1, 2), (3, 1, -2)\}$ is a linearly independent subset of $\mathbb{R}^3(\mathbb{R})$.

Suppose $a, b, c \in \mathbb{R}$ such that

$$a(2, 1, 4) + b(1, -1, 2) + c(3, 1, -2) = 0$$

$$\Rightarrow 2a + b + 3c = 0; a - b + c = 0; 4a + 2b - 2c = 0$$

$$\text{i.e., } \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 4 & 2 & -2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\xrightarrow[R_3 - 2R_1]{R_2 - 2R_1} \begin{bmatrix} 2 & 1 & 3 \\ 0 & -4 & -1 \\ 0 & 0 & -8 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Therefore, we get $a = 0, b = 0, c = 0$

Hence the vectors $(2, 1, 4), (1, -1, 2), (3, 1, -2)$ are linearly independent.

② Show that the set $S = \{(1, 2, 1), (3, 1, 5), (3, -4, 7)\}$ is a linearly dependent subset of $\mathbb{R}^3(\mathbb{R})$.

Note: If every finite subset of an infinite set S of vectors is linearly independent, then S is linearly independent.

(2) If $S = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is a linearly independent subset of a vector space $V(F)$, then every element $\alpha \in L(S)$ has a unique representation in the form $a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n$, where $a_1, a_2, \dots, a_n \in F$.

Basis of a vector space:

Definition: Let $V(F)$ be a vector space and S any non-empty subset of V . Then S is said to be basis of V if (i) $(\text{all } \mathbb{R})$ S is linearly independent
(ii) $L(S) = V$

Examples: (1) $S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$

is linearly independent and $L(S) = \mathbb{R}^3(\mathbb{R})$.

$\therefore S$ is a basis of $\mathbb{R}^3(\mathbb{R})$.

This basis is called ~~the~~ standard basis of $\mathbb{R}^3(\mathbb{R})$.

(2) $S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

is the standard basis of $M_{2 \times 2}(\mathbb{R})$

③ verify the vectors $(1, 2, 1)$, $(3, 1, 5)$, $(3, -4, 7)$ form a basis of $\mathbb{R}^3(\mathbb{R})$ or not.

Sol: Let $a, b, c \in \mathbb{R}$ such that

$$a(1, 2, 1) + b(3, 1, 5) + c(3, -4, 7) = 0$$

$$\begin{aligned}\text{Then } a + 3b + 3c &= 0 \\ 2a + b - 4c &= 0 \\ a + 5b + 7c &= 0\end{aligned}$$

$$\text{i.e., } \begin{bmatrix} 1 & 3 & 3 \\ 2 & 1 & -4 \\ 1 & 5 & 7 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\xrightarrow[R_3 - R_1]{R_2 - 2R_1} \begin{bmatrix} 1 & 3 & 3 \\ 0 & -5 & -10 \\ 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\xrightarrow{5R_3 + 2R_2} \begin{bmatrix} 1 & 3 & 3 \\ 0 & -5 & -10 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Therefore, a, b, c are all not zero ~~and hence~~.

Hence, the vectors $(1, 2, 1)$, $(3, 1, 5)$, $(3, -4, 7)$ are linearly dependent.

Thus, the vectors $(1, 2, 1)$, $(3, 1, 5)$, $(3, -4, 7)$ do not form a basis of $\mathbb{R}^3(\mathbb{R})$.

Dimension of a vector space:

Let $V(F)$ be a vector space. The number of elements of a basis of V is called the dimension of V over F and we write $\dim_F(V)$ or simply $\dim(V)$.

4.2

Finite dimensional vector space

Def: Let V be a vector space over a field F . Then we say that V is finite dimensional if it is spanned by a finite set of vectors.

Def: The dimension of a finite dimensional vector space V is the number of elements in a basis of V .

Note: (1) For convenience, we will define the dimension of the trivial vector space $\{0\}$ to be zero, even though $\{0\}$ doesn't have a basis.

(2) Every basis of an n -dimensional vector space contains exactly n vectors.

(3) Let $V(F)$ be an n -dimensional vector space and $S = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be a subset of V . Then

(i) If S is linearly independent, then S is a basis of V .

(ii) If $L(S) = V$, then S is a basis of V .

Examples: (1) The dimension of a vector space $\mathbb{R}(\mathbb{R})$ is one.

② The dimension of a vector space $\mathbb{C}(\mathbb{R})$ is two.

③ For any positive integer n , let P_n denote the vector space of polynomials with real coefficients of degree at most n . Then the set $S = \{1, x, x^2, x^3\}$ is a basis for P_3 . Hence the dimension of P_3 is 4.

Note: ① Let $V(F)$ be a vector space. The set $S = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ of vectors in V form a basis of V if and only if every vector α in V can be expressed uniquely as a linear combination of $\alpha_1, \alpha_2, \dots, \alpha_n$.

② Let $V(F)$ be an n -dimensional vector space. Then any subset of V contains more than n elements is linearly dependent.

③ The set of vectors consisting of a zero vector is linearly dependent.

④ If $\alpha = a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n$, then the set $\{\alpha, \alpha_1, \alpha_2, \dots, \alpha_n\}$ is linearly dependent.

Problems:

① Find a basis and dimension of the subspace

$w = \{(x, y, z, w) \in \mathbb{R}^4 \mid x - y + z - w = 0\}$ of a vector space $\mathbb{R}^4(\mathbb{R})$.

Sol: Any vector $\alpha = (x, y, z, w)$ in w satisfies $x - y + z - w = 0$, or equivalently

$$x = y - z + w.$$

Therefore, we have

$$\begin{aligned}(x, y, z, w) &= (y - z + w, y, z, w) \\ &= y(1, 1, 0, 0) \\ &\quad + z(-1, 0, 1, 0) \\ &\quad + w(1, 0, 0, 1)\end{aligned}$$

and hence, the set $S = \{(1, 1, 0, 0), (-1, 0, 1, 0), (1, 0, 0, 1)\}$

is a spanning set for the subspace w .

$$\text{i.e., } L(S) = w.$$

Suppose $a, b, c \in \mathbb{R}$ such that

$$a(1, 1, 0, 0) + b(-1, 0, 1, 0) + c(1, 0, 0, 1) = 0$$

$$\begin{aligned}\text{Then } a - b + c &= 0; \quad a = 0, \quad b = 0, \quad c = 0 \\ &\Rightarrow a = 0, \quad b = 0, \quad c = 0.\end{aligned}$$

Therefore, $S = \{(1, 1, 0, 0), (-1, 0, 1, 0), (1, 0, 0, 1)\}$
is linearly independent.

Hence $S = \{(1, 1, 0, 0), (-1, 0, 1, 0), (1, 0, 0, 1)\}$
is a basis for the subspace W .

Thus $\dim W = 3$.

- (2) Find the basis and dimension of the subspace $W = \{(x, y, z) \mid x - y = 0\}$ of a vector space $\mathbb{R}^3(\mathbb{R})$.

Answer: Basis is $\{(1, 1, 0), (0, 0, 1)\}$
and dimension 2.

- (3) Show that the set $S = \{(2, 1, 4), (1, -1, 2), (3, 1, -2)\}$ form a basis for a vector space $\mathbb{R}^3(\mathbb{R})$.

- (4) Is the set $S = \{(1, 2, 1), (2, 1, 0), (0, 0, 1)\}$ form a basis for $\mathbb{R}^3(\mathbb{R})$.

4.3 Row Space, Column Space and Nullspace.

Terminology: Let A be the 2×3 matrix $\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \end{bmatrix}$.

Then (i) The row vectors of A are $(1, 2, -1)$
 $[1 \ 2 \ -1]$ and $[0 \ 1 \ 3]$ in $\mathbb{R}^3$

(ii) The column vectors of A are
 $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \end{bmatrix}$ in $\mathbb{R}^2$.

Definition: Let A be a $m \times n$ matrix. Then

(i) The row space of A is the subspace of $\mathbb{R}^n$ spanned by the row vectors of A .

(ii) The column space of A is the subspace of $\mathbb{R}^m$ spanned by the column vectors of A .

Note: If a matrix A is row-equivalent to a matrix B in row-echelon form, then the non-zero row vectors of B form a basis for the row space of A .

Example: Finding a basis for Row space.

(i) Let $A = \begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 1 & -1 & 0 & 0 & 2 \\ 2 & 1 & 6 & 0 & 1 \end{bmatrix}$, then find a basis for row space of A .

$$\text{Sol: } A = \begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 1 & -1 & 0 & 0 & 2 \\ 2 & 1 & 6 & 0 & 1 \end{bmatrix} \xrightarrow{\substack{R_4 - R_1 \\ R_5 - 2R_1}} \begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & -2 & -4 & -1 & 0 \\ 0 & -1 & -2 & -2 & -3 \end{bmatrix}$$

$$\xrightarrow{\substack{R_4 + 2R_2 \\ R_5 + R_2}} \begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & -1 & -2 \end{bmatrix}$$

$$\xrightarrow{\substack{R_4 - R_3 \\ R_5 + R_3}} \begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(Echelon form of A)

Therefore $\alpha_1 = (1, 1, 4, 1, 2)$, $\alpha_2 = (0, 1, 2, 1, 1)$

$\alpha_3 = (0, 0, 0, 1, 2)$ form a basis for the row space of A. Hence the dimension of the row space (of A) is 3.

Example: Finding a basis for the column space of A.

Note: A basis for the row space of A^T is the same as finding a basis for the column space of A.

① Let $A = \begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 1 & -1 & 0 & 0 & 2 \\ 2 & 1 & 6 & 0 & 1 \end{bmatrix}$, then find a basis for the ~~row~~^{column} space of A .

Sol: $A^T = \begin{bmatrix} 1 & 0 & 0 & 1 & 2 \\ 1 & 1 & 0 & -1 & 1 \\ 4 & 2 & 0 & 0 & 6 \\ 1 & 1 & 1 & 0 & 0 \\ 2 & 1 & 2 & 2 & 1 \end{bmatrix}$

$\xrightarrow{\substack{R_2 - R_1; R_3 - 4R_1 \\ R_4 - R_1; R_5 - 2R_1}}$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & -2 & -1 \\ 0 & 2 & 0 & -4 & -2 \\ 0 & 1 & 1 & -1 & -2 \\ 0 & 1 & 2 & 0 & -3 \end{bmatrix}$$

$\xrightarrow{\substack{R_3 - 2R_2 \\ R_4 - R_2; R_5 - R_2}}$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & -2 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 2 & 2 & -2 \end{bmatrix}$$

$\xrightarrow{R_3, 4}$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & -2 & -1 \\ 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 2 & -2 \end{bmatrix}$$

$\xrightarrow{R_5 - 2R_3}$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & -2 & -1 \\ 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(Echelon form of A^T)

Therefore, $\alpha_1 = (1, 0, 0, 1, 2)$, $\alpha_2 = (0, 1, 0, -2, -1)$, $\alpha_3 = (0, 0, 1, 1, 1)$ form a basis for the column space of A . Hence, the dimension of the column space is 3.

Note: If A is an $m \times n$ matrix, then the row space and the column space of A have the same dimension.

Definition: The dimension of the row (or, column) space of a matrix A is called the rank of A . It is denoted by $\text{rank}(A)$ (or, $\rho(A)$).

Definition: If A is a $m \times n$ matrix, then the set of all solutions of the homogeneous system of linear equations $Ax = 0$ is a subspace of $\mathbb{R}^n$ called the nullspace of A and denoted by $N(A)$.

$$\text{i.e., } N(A) = \left\{ x \in \mathbb{R}^n \mid Ax = 0 \right\}.$$

The dimension $N(A)$ is called the nullity of A .

Example: Finding a basis for the nullspace of A.

① let $A = \begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 1 & -1 & 0 & 0 & 2 \\ 2 & 1 & 6 & 0 & 1 \end{bmatrix}$, then find a basis and dimension for $N(A)$.

Sol: we need to solve the system $Ax = 0$

i.e., $\begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 1 & -1 & 0 & 0 & 2 \\ 2 & 1 & 6 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$\xrightarrow[R_5 - 2R_1]{R_4 - R_1} \begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & -2 & -4 & -1 & 0 \\ 0 & -1 & -2 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$\xrightarrow[R_5 + R_2]{R_4 + 2R_1} \begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & -1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$\xrightarrow[R_5 + R_3]{R_4 - R_3} \begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$$\xrightarrow{R_1 - R_2} \begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\xrightarrow{R_2 - R_3} \begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & 2 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Therefore, we get

$$x_1 + 2x_3 + x_5 = 0$$

$$x_2 + 2x_3 - x_5 = 0$$

$$x_4 + 2x_5 = 0.$$

Let $x_5 = k_1$ and $x_3 = k_2$, then

$$x_1 = -k_1 - 2k_2$$

$$x_2 = k_1 - 2k_2$$

$$x_3 = k_2$$

$$x_4 = -2k_1$$

$$x_5 = k_1$$

$$\therefore x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -k_1 - 2k_2 \\ k_1 - 2k_2 \\ k_2 \\ -2k_1 \\ k_1 \end{bmatrix} = k_1 \begin{bmatrix} -1 \\ 1 \\ 0 \\ -2 \\ 1 \end{bmatrix} + k_2 \begin{bmatrix} -2 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

Hence, the set $\left\{ \begin{bmatrix} -1 \\ 0 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ -2 \\ 1 \\ 0 \end{bmatrix} \right\}$ form a basis for $N(A)$.

Thus $\dim(N(A)) = 2$. i.e., $\text{nullity}(A) = 2$.

Note: If A is a $m \times n$ matrix of rank r , then the dimension of the solution space of $Ax = 0$ is $n - r$.

i.e., $\dim(N(A)) = n - \text{rank}(A)$

or, $\boxed{\text{rank}(A) + \text{nullity}(A) = n}$
 $\downarrow$
 (number of ~~rows~~ columns of A .)

Problems

① Let $A = \begin{bmatrix} 2 & 4 & -3 & -6 \\ 7 & 14 & -6 & -3 \\ -2 & -4 & 1 & -2 \\ 2 & 4 & -2 & -2 \end{bmatrix}$.

- Find (i) basis for ~~row~~ row space of A
 (ii) basis for column space of A
 (iii) basis for nullspace of A
 (iv) $\text{rank}(A)$ (v) $\text{nullity}(A)$.

Answer: (i) $\{(1, 2, -1, -1), (0, 0, 1, 4)\}$
 (ii) $\{(2, 7, -2, 2), (-3, -6, 1, -2)\}$
 (iii) $\left\{ \begin{bmatrix} -1 \\ 0 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ -2 \\ 1 \\ 0 \end{bmatrix} \right\}$ (iv) 2 (v) 2.

Practice all the questions

1. Is the set $S = \{(1, 2, 1), (3, 1, 5), (3, -4, 7)\}$ linearly independent?

Answer: The set $S = \{(1, 2, 1), (3, 1, 5), (3, -4, 7)\}$ is linearly dependent

2. Is the set $S = \{(2, 1, 4), (1, -1, 2), (3, 1, -2)\}$ forms a basis of a vector space $\mathbb{R}^3(\mathbb{R})$.

Answer: Yes, S is a basis of $\mathbb{R}^3(\mathbb{R})$.

3. If W is the subspace of a vector space $\mathbb{R}^4(\mathbb{R})$ generated by the vectors $(1, -2, 5, -3)$, $(2, 3, 1, -4)$, $(3, 8, -3, -5)$, then find a basis and dimension of W .

$$\text{Answer: } \begin{pmatrix} 1 & -2 & 5 & -3 \\ 2 & 3 & 1 & -4 \\ 3 & 8 & -3 & -5 \end{pmatrix} \xrightarrow[\substack{R_2-2R_1 \\ R_3-3R_1}]{R_2-2R_1} \begin{pmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 14 & -18 & 4 \end{pmatrix} \xrightarrow{R_3-2R_2} \begin{pmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Therefore, the two non-zero row vectors $(1, -2, 5, -3)$, $(0, 7, -9, 2)$ forms a basis of W and hence $\dim W=2$.

4. Find the basis and dimension of the subspace $W = \{(x, y, z, w) \mid x+y=0, z=2w\}$ of a vector space $\mathbb{R}^4(\mathbb{R})$.

Answer: The set $\{(-1, 1, 0, 0), (0, 0, 2, 1)\}$ is a basis of W and hence $\dim W=2$.

5. Find the basis and dimension of the row space of $A = \begin{pmatrix} 1 & -2 & 5 & -3 \\ 2 & 3 & 1 & -4 \\ 3 & 8 & -3 & -5 \end{pmatrix}$.

Also find the nullity (A).

Answer: The set $\{(1, -2, 5, -3), (0, 7, -9, 2)\}$ form a basis for the row space of A and hence the dimension of the row space of A is 2(i.e., the rank of A is 2). Also nullity (A) = $4-2 = 2$.